

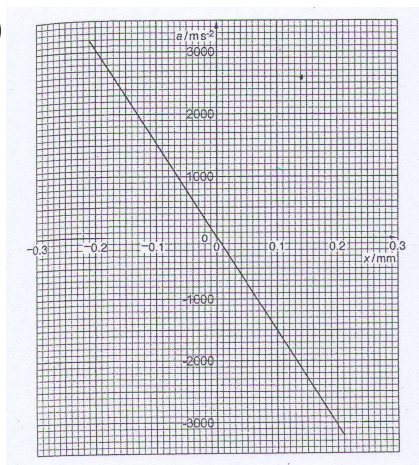
Paper 3

- 1 (a) Simple harmonic motion is defined as an oscillatory motion taking place in which the acceleration of an object is directly proportional to the displacement of the object from its equilibrium position, and the acceleration is always directed towards that position.

(b) (i) 1. $\omega = 2\pi f = 2\pi(620) = 3.9 \times 10^3 \text{ rad s}^{-1}$

2. $a_0 = \omega^2 x_0 = (3896)^2 (0.21 \times 10^{-3}) = 3.2 \times 10^3 \text{ m s}^{-2}$

(ii)



*A scale for the vertical axis must be given.

- (c) If the frequency of the sound produced matches the natural frequency of the cone of the loudspeaker, resonance will occur, causing the cone to vibrate at maximum amplitude. As a result, the frequency response of the loudspeaker will no longer be uniform, as it will vibrate more at certain frequencies and less at others. The sound output will be distorted.

2 (a) (i) $P = \frac{V}{R} = \frac{4.5}{1.2} = 3.75 \text{ W}$